

Managing Drone Traffic in Urban Skyway Networks

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Abstract

We propose a novel drone traffic monitoring system for smart cities, featuring skyway networks that connect rooftop hubs equipped with recharging stations for long-distance deliveries. However, *inter-drone interference* may impact the *safe* and *efficient* provisioning of drone delivery services as drones operate in shared aerial pathways and utilize recharging stations. To address this challenge, we propose a framework to evaluate the impact of interferences on drones' delivery *efficiency* and *safety*, accompanied by a purpose-built tool implementing the framework. The tool is based on real data gathered through drones' flights in an indoor testbed. The insights generated by the tool allow for optimizing drones' flight paths by mitigating and resolving potential inter-drone interferences.

CCS Concepts

• Human-centered computing → Ubiquitous and mobile computing.

Keywords

Drones, UAVs, Drone Deliveries

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1 Introduction

The growth of e-commerce has heightened demand for efficient package delivery, positioning drones as a leading solution. Logistics companies worldwide are adopting drone-based services for goods and medical supply transport [4]. Recent research on *service-based drone delivery* focuses on integrating UAVs into the service paradigm, with drones serving as the primary delivery agents [3]. These drones usually operate in a *skyway network*, an interconnected set of *nodes*. The nodes are building rooftops that serve as *recharging*

stations or delivery destinations. These *nodes* are linked by Line-of-Sight (LoS) *skyway segments* which provide designated airspace corridors for drone transit. As the number of drones increases in a skyway network, the complexity of managing these operations grows, particularly in minimizing *inter-drone interference*.

Recent work in service-based drone deliveries focuses on composing delivery paths within the skyway network. These methods consider dynamic weather conditions and drones' constraints (e.g., battery capacity and flight range) to serve delivery requests from the source to the destination node [8]. These approaches allow *drone providers* (i.e., drone owners) to compose optimal paths consisting of segments and nodes, tailored to their specific operational goals. However, *inter-drone interferences* may occur when multiple drone providers compose their trajectory plan independently and *selfishly* to achieve their goals. This interference can negatively impact *safe* and *efficient* provisioning of drone providers' Quality of Service (QoS) requirements. For example, turbulence from nearby drones operating in the same segment can destabilize flight, forcing a drone to use more energy to maintain its trajectory and altitude. This increased energy consumption reduces efficiency and violates the provider's expected battery usage [9]. This underscores the critical need for effective drone traffic management in the skyway network to ensure providers' QoS expectations are consistently met. Conducting large-scale experiments in real-world skyway networks is restricted by government regulations due to safety concerns and the complexity of managing multiple drones in an outdoor environment [6]. Therefore, there is a need for a smart strategy that not only emulates drone flights but also enables detailed analysis of interferences in multi-drone skyway networks.

Traditional simulation tools are often limited to 2D scenarios, restricting their ability to capture the intricate dynamics of drone traffic within a 3D airspace [1, 2]. Existing drone traffic management simulators mainly focus on ensuring safety, particularly by preventing collisions between drones. To the best of our knowledge, no prior frameworks have been developed to analyze the *interference* between drones at recharging stations or during segment traversals. Specifically, these approaches do not account for how interference at recharging stations or during segment traversals may disrupt service quality, operational efficiency, or the timely delivery of goods.

We define *interference* as an event that impacts the fulfillment of drone providers' QoS requirements to achieve the desired *delivery time*. These *interferences* can occur at the *node-level*, where drones queue to recharge, or at the *segment-level*, where drones traverse aerial pathways. At the node-level, interference may cause drones to



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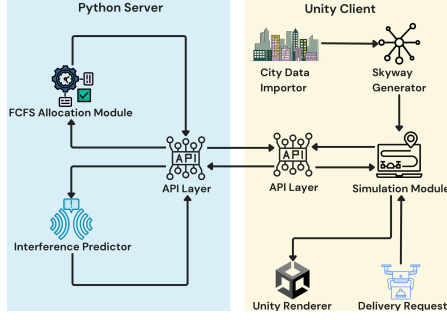


Figure 1: System Architecture

hover due to unavailable recharging pads, resulting in increased battery consumption, extended recharging times, and delivery delays. In contrast, *segment-level interference* may occur as drones operate in close proximity while traversing through segments, causing aerodynamic forces to interfere. This can alter the anticipated power consumption, requiring more recharging at subsequent nodes. For instance, the turbulent air generated by a drone may destabilize nearby drones, causing them to use additional battery power to maintain stability [9].

We propose a 3D drone traffic monitoring system to model and analyze the impact of *interferences* on drone delivery efficiency in a multi-drone skyway network. The skyway network is based on a realistic urban layout of the Sydney CBD area. This system leverages a preliminary interference model developed from real-world drone flights conducted in a small-scale indoor drone testbed [9]. We extend this model for large scale drones used in real-world delivery services to emulate dynamics of drone delivery in realistic urban conditions with varying wind conditions. Insights from the 3D drone traffic monitoring system allows the drone traffic management authority to devise strategies for mitigating or eliminating interference impacts. This optimization would allow drone providers to meet their QoS requirements, particularly in terms of delivery time.

2 Demo Setup

This demo paper presents a *drone traffic monitoring system* that serves as a *pre-requisite* for ensuring the *safe* and *efficient* operation of drone delivery services within a *skyway network*, by detecting and assessing inter-drone interference. The system's computations are based on an interference model developed through real-world experiments. It evaluates the dynamics of drone delivery flights, accounting for interferences in shared skyway segments due to overlapping flight paths, and at nodes, where limited recharging resources may cause delays. It quantifies the impact of interference on meeting QoS requirements, particularly regarding delivery time. This evaluation is crucial for mitigating interferences and ensuring *seamless* drone traffic within the skyway network.

2.1 System Architecture

The proposed system architecture integrates a *Python Server* and a *Unity Client* (Fig. 1). The *Python Server* predicts *interference* events and evaluates their impact on drones' *energy consumption* in shared

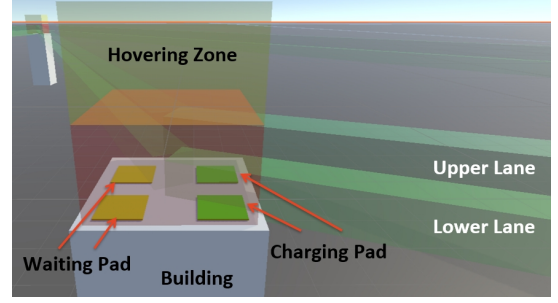


Figure 2: Node and Segment Configuration



Figure 3: Flight paths of drones in a skyway network

skyway segments. It also computes waiting times and energy consumption due to hovering or waiting at nodes, based on a First-Come-First-Served (FCFS) approach for recharging pad allocation. The *Unity Client* constructs the skyway network based on urban geographical data. This geographical data consists of various attributes such as GPS coordinates, building shape, type, height, and rooftop area. Furthermore, the *Unity Client* also generates weather conditions (i.e., wind speed and wind direction) for the urban area. Data from the *Unity Client* is then sent to the server in real-time to enable accurate modeling of inter-drone interferences.

2.1.1 Skyway Network. The environment presented in this paper is based on the city infrastructure of the Sydney CBD area. The proposed system is designed to implement the skyway network for any geographical area, allowing for customization based on specific urban infrastructures. In this case, we have used the Sydney CBD area as a model, but the system can easily adapt to different cities or regions by incorporating their unique geographical and infrastructural data. Nodes are equipped with a fixed number of charging and waiting pads, determined by the rooftop capacity of buildings and peak-time drone traffic demand. Additionally, each node features a designated *hovering zone*, where drones can wait in a circular pattern for their turn to access the waiting or recharging pads (Fig. 2). Nodes are interconnected through Line-of-Sight (LoS) *skyway segments*, which act as aerial corridors with pre-defined 3D volumes, in compliance with CASA's drone safety regulations. Each segment comprises two lanes for bidirectional drone traffic, ensuring safe operations between nodes (Fig. 2). Moreover, the 3D volume of each segment extends vertically from the building rooftop up to a maximum height of 400 feet, in compliance with CASA's drone flying regulations [5].

2.1.2 Multi-Drone Traffic Emulation. The framework accommodates requests from multiple drone service providers. Each request includes the segments to be traversed by the drone from the source to the destination node, as well as the recharging duration at intermediate nodes. The drone provider’s path composition aims to minimize cost, *time*, or *energy consumption* [8]. Figure 3 illustrates the flight paths composed by multiple drone providers.

Drones may stop at intermediate nodes to recharge as per the delivery plan composed by the drone provider. However, this plan does not fully account for the arrival and departure of other drones at these nodes. In real-time, drones may get delayed due to the limited availability of recharging pads on arrival. The proposed traffic monitoring system considers the arrival and departure of all drones at nodes to identify delays provoked by the simultaneous arrivals of drones at a node. A drone may experience one of three states upon arrival at a node: *direct recharging* at a recharging pad, *waiting* at a waiting pad, or *hovering* in a designated hovering zone based on the FCFS approach. Figure 4 illustrates a scenario where three drones, arriving 10 seconds apart, require recharging at a node with one recharging pad and one waiting pad.

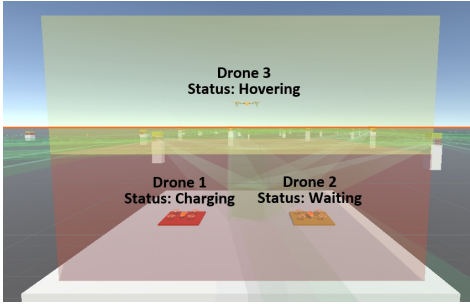


Figure 4: The node-level interference between drones

The framework allocates recharging pads based on the FCFS principle. Therefore, Drone 3 hovers in the hovering zone until Drone 1 and Drone 2 finish recharging. The framework captures delivery delays caused by *waiting* and *hovering* times due to simultaneous drone arrivals at nodes. Table 1 shows an instance of the output from a CSV log file for the aforementioned scenario. The framework determines each drone’s actual departure time from the node due to interference from other drones. We can observe that the Expected Depart Time (i.e., the scheduled time a drone is expected to leave after charging) and the Actual Depart Time (i.e., the time the drone actually departs due to unavailability of recharging pad) are different for Drone 2 and Drone 3. For example, Drone 3 expects to depart at 11:59 AM after ≈ 42 minutes of recharging, on arrival at 11:16 AM. However, due to the queuing, its departure is delayed to 12:41 PM. This delay is more significant for Drone 3, as hovering leads to an additional battery power consumption, thus resulting in a longer recharging time than initially anticipated by the provider.

We use Kirschstein’s energy consumption model [7] to compute the power consumption of drones as they traverse the segment. This model considers various factors such as drones’ mass, payload, flight angle, and speed for computing energy consumption. However, the proximity of a drone to its peer drones significantly

Table 1: The queuing behavior of drones at a node

Drone ID	Arrival Time	Charging Time (min)	Waiting Time (min)	Hovering Time (min)	Expected Depart Time	Actual Depart Time
1	10:46	36.52	0	0	11:22 AM	11:22 AM
2	10:56	36.52	25.56	0	11:33 AM	11:59 AM
3	11:16	42.69	36.52	6.52	11:59 AM	12:41 PM

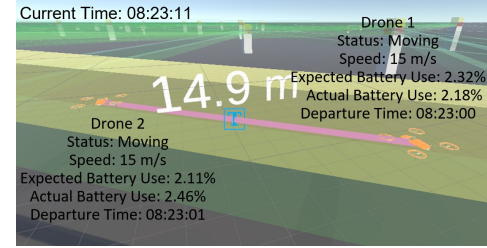


Figure 5: Segment-level interference between drones

affects its energy consumption [9]. This is due to the interaction of the aerodynamic forces between drones as they operate in close proximity in a segment. For instance, downwash forces from a higher-altitude drone can destabilize a lower-altitude drone, causing it to consume more battery power to maintain stability. The proposed framework utilizes an interference model to calculate the additional energy consumption of drones caused by interference from concurrently operating peer drones within the segment [9]. However, this interference model is designed for small-scale drones Crazyflie 2.1¹, which are not suitable for outdoors due to their compact size. We leverage this preliminary model for the drones used in real-world scenarios for delivering goods. In this regard, we train the polynomial regression model specifically for the DJI FlyCart 30². The proposed energy consumption model computes the impact on the initially anticipated energy consumption of the drone based on the payload, position, proximity to the peer drone, and dynamic wind conditions.

The proposed tool enables real-time visualization of the impact of inter-drone interference as drones traverse a segment. It tracks the spatial separation between drones sharing the segment every minute, using the proposed model to compute the impact on the resulting energy consumption. Figure 5 illustrates an instance of a drone flight scenario from the drone traffic monitoring system. In this scenario, two drones are operating in a skyway segment at the same altitude along the x -axis, with Drone 2 positioned directly behind Drone 1. The drones have a longitudinal separation of 14.9 m along the x -axis over a 10-second interval, leading to a 17% increase in power consumption for Drone 2. This additional energy use is due to the instability caused by the turbulent airflow from Drone 1’s propellers, which increases the drag force on Drone 2, requiring it to expend extra energy to maintain stable flight.

¹<https://www.bitcraze.io/products/old-products/crazyflie-2-1/>

²<https://www.dji.com/au/flycart-30>

2.2 Data Collection

The proposed drone traffic monitoring system comprehensively collects the data for each drone delivery flight. It tracks the drones' flight path, including all skyway segments and nodes visited by the drones. For each skyway segment, the system collects the traversal time, 3D position, battery level, and 3D spatial separation for all drones operating within the segment. For each node, the system logs the arrival and departure times of drones, along with the waiting and hovering durations caused by the unavailability of charging pads. There are two main key features of the proposed system; (1) It provides dynamic visualization of interferences occurring at *nodes* and *segments* based on real-world conditions (2) It generates detailed CSV log files that offer insights into the interferences' location and impact on fulfilling the drone providers' QoS requirements, particularly in terms of delivery time. These insights allow the drone traffic management authority to optimize the skyway network infrastructure and optimize drone delivery paths by resolving inter-drone interferences.

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